



Practice Question

Profit & Loss

1. What will come in place of question mark (?) in the following questions?

C.P. (Rs.)	Profit		Loss		S.P (Rs.)
	(Rs.)	(%)	(Rs.)	(%)	
500	200				?
500		20%			?
500			200		?
500				20%	?
?	200				500
?		20%			600
?			200		600
?				20%	400
500	?	?			600
600			?	?	540

2. Shiv buys a computer for Rs. 39,000 and sells it for Rs. 21,450/-. Find his Loss%?

3. Ritesh sells his bike for Rs. 13,050 and earned 45% profit. What would be the % profit / loss, if 15 such bikes were sold for Rs. 1,64,700?

4. Raghvan sells his bike for Rs 1, 37,750 and fetched 45% profit. Had he sold it for Rs 52,250 what would have been his loss percent?

5. If the C. P. is 60% of the S.P. what would be the Profit %?

6. If the S. P. is 65% of the C.P. what would be the Loss %?
7. When a seller sells his article for Rs. 32,060 , he incurs 30% loss on the transaction. At what price should he sell the article to earn 10% profit on the transaction ?
8. If the C.P. of 35 article is equal of S.P. of 28 article. Find the gain or loss %.
9. I sell 35 articles for the same money as I paid for 42 articles. What is my gain / loss %?
10. If the C.P. of 32 article is equal of S.P. of 40 article. Find the gain or loss %.
11. I sell 25 articles for the same money as I paid for 15 articles. What is my gain / loss %.
12. The cost price of a car is one fifth of its selling price. What is the gain/loss percent on the transaction ?
13. Ramendra purchased a bike for Rs 72,500 and a laptop for Rs 32,000. He sold the bike at a profit of 20% and the laptop at a loss of 25%. Find his overall loss or profit percentage on the whole transaction?
14. Nikhil purchased a house for Rs. 20,00,000 and gave it on rent immediately at the rate of Rs. 15,000 per month. After five months, he sold the house for Rs. 23,00,000. Find his total profit % earned in the transaction.
15. A vendor purchased 60 kg of apples for Rs. 4,500 and sold the whole lot at the rate of Rs. 80 per kg. What will be his profit %?
16. The ratio between the sale price and the cost price of an article is 32: 28. What is the ratio between the profit and the cost price of that article?

17. The profit earned by selling an article for Rs.5,200 is equal to the loss incurred when the same article is sold for Rs.4,500. What should be the sale price for making 15% profit?
18. A man bought 18 glasses for Rs.12 and sold 12 glasses for Rs 18. What is the profit percent?
19. A man bought boxes at the rate of 30 for Rs. 25 and sold them at the rate of 25 for Rs.30. Calculate no. of boxes he should sell to earn a profit of Rs. 770.
20. A shopkeeper earns a profit of 30% by selling an article. What would be the percent change in the profit percent, if he had paid 20% less and sold at 20% more?
21. If a shopkeeper buys toys at a prices ranging from Rs. 175 to Rs. 200 and sells at prices ranging from Rs. 275 to Rs. 300, what is the greatest possible profit that might be made in selling 45 toys?
22. If on selling 20 articles, a seller makes a profit equal to the selling price of 8 articles, what is his percent profit?
23. A shopkeeper sells two items at the same price for Rs 2800 each. If he sells one item at 40% profit and other at 30% loss, find the loss incurred by the shopkeeper in the whole transaction.
24. A Shopkeeper sold an article at 5% loss. If he had sold it for Rs. 350 more, he would have made 20% profit, find the selling price of the article?
25. A Shopkeeper sold an article at 20% loss. If he had sold it for Rs. 420 more, he would have made 40% profit, find the selling price of the article?
26. When a shopkeeper sells 6 articles for Rs 30, he earns a profit of 25%. What will be his gain or loss percent if he sells 14 articles at Rs 61.60?

27. A shopkeeper sells two items at the same price for Rs 250 each. If he sells one item at 10% profit and other at 10% loss, find the loss incurred by the shopkeeper in the whole transaction.
28. Dushyant bought one computer and one printer for Rs 31,200. If the cost of computer is 60% more than the printer what is the difference between the price of computer and printer?
29. A shopkeeper sells his goods at C.P. but uses a weight of 850 gms. Find his gain %.
30. A dishonest seller uses a weight of 900 gms instead of 1 kg and also sells his goods at 30% more than CP. Find his approximate profit percent.
31. A dishonest seller uses a weight of 900 gms instead of 1 kg and also sells his goods at 10% more than CP. Find his approx. profit percent.
32. A dishonest seller, at the time of selling and purchasing uses weight 10% less and 20% more per kg respectively. Find the approx. percent profit earned by him if he sells his goods at cost price.
33. A motorbike marked at Rs 22,500 is sold for Rs 18,000. Find the rate of discount ?
34. A manufacturer offers a 30% rebate on the marked price of a product. The retailer offers another 40% rebate on the reduced price. The two reductions are equivalent to a single reduction of:
35. A tradesman marks his goods 20% above the C.P. If he allows a discount of 10 %, then what is his profit % ?
36. After successive discounts of 30% and 20%, an article was sold for Rs 56,000. What was the original price of the article?

37. After successive discounts of 14% and 8%, an article was sold for Rs 31,648. What was the original price of the article?
38. The price of a computer is marked at Rs 36,000. If successive discounts of 15%, 10% and 5% allowed, then at what price does a customer buy it?
39. A shopkeeper gives 15% additional discount on the discounted price, after giving an initial discount of 25% on the labeled price of a radio. If the final sale price of the radio is Rs 7650, then what is its labeled price?
40. If the S.P. of Rs 2,250 results in a 10% discount on list price, what S.P. would result in a 20% discount on list price?
41. Marked price of an article at a shop is Rs. 3,000. If it is being sold at successive discounts of 15% and 20%, what would be its net selling price ?
42. After getting two successive discounts, a computer with a list price of Rs 20,000 is available at Rs 12,000. If the second discount is 20%, find the first discount.
43. The price of a printer is marked at Rs 15,000. If successive discounts of 10%, 20% and 25% allowed, then at what price does a customer buy it?
44. A trader marks all his goods at 60% above the costs price and offers a discount of 15% on the marked price. What is the actual profit % on the sales?
45. If a company sells a bikes with a marked price of Rs 3,50,000 and gives a discount of 3% on Rs 1,80,000 and 5% on the remaining amount of Rs 1,70,000, then what is the actual price charged by the company for the bike ?

46. Girish purchased a car with an additional 20% discount on the reduced price after deducting 40% on the labeled price. If the labeled price was Rs 5,50,000, at what price did he purchase the car?

47. Sneha bought a music system with 15% discount on the labeled price. Had she bought it with 20% discount, she would have saved Rs. 500. At what price did he buy the cupboard?

48. After giving a discount of 20% on the marked price of a TV, a shopkeeper earns a profit of 25% on the cost price. Had he not given any discount on the marked price, what would have been the profit percent earned by him?

49. A Shopkeeper marks the price of a refrigerator at Rs. 30,000/- and gives a discount of 15%. He also gives a mixer grinder worth Rs. 1,500 free with the refrigerator and actually earns a profit of 20% by selling the refrigerator. Find the cost price of the refrigerator.

50. A Shopkeeper marks the price of a mobile phone at Rs. 6,000/- and gives a discount of 10%. He also gives a pen drive worth Rs. 600 free with the mobile phone and actually earns a profit of 25% by selling the mobile phone. Find the cost price of the phone.

Solution

Sol-1

C.P. (Rs.)	Profit		Loss		S.P (Rs.)
	(Rs.)	(%)	(Rs.)	(%)	
500	200	-	-	-	$500 + 200 = 700$
500	-	20%	-	-	$120\% \text{ of } 500 = 600$
500	-	-	200	-	$500 - 200 = 300$
500	-	-	-	20%	$80\% \text{ of } 500 = 400$
$500 - 200 = 300$	200	-	-	-	500
$120\% \text{ of C.P.} = 600$ $\Rightarrow \text{C.P.} = 500$	-	20%	-	-	600
$600 + 200 = 800$	-	-	200	-	600
$80\% \text{ of C.P.} = 400$ $\Rightarrow \text{C.P.} = 500$	-	-	-	20%	400
500	100	20%	-	-	600
600	-	-	60	10%	540

Sol-2

$$\text{Loss} = 39,000 - 21,450 = 17,550 \text{ Rs.}$$

$$\% \text{ Loss} = (17550/39000) \times 100 = 45\%$$

Sol-3

S.P of the bike = Rs 13050

Profit = 45 %

Let C.P. of 1 bike = Rs. x

$$x \times 145\% = 13,050$$

$$x = (13050 / 145) \times 100$$

$$x = 9,000$$

C.P. of 1 bike = Rs. 9,000

Now, S.P of 1 bike = $164700/15 = 10980$

Now, S.P of 1 bike = $(10980 - 9000) / 9000 \times 100$

S.P of 1 bike = $1980 / 9000 \times 100 = 22\%$ profit

Sol-4

Let C.P of the bike = Rs. x

$$\therefore x \times 145\% = 1,37,750$$

$$\Rightarrow x = (137750 \times 100) / 145 = \text{Rs. } 95,000$$

Assumed S.P = Rs 52,250

$$\text{Loss \%} = (95000 - 52250) / 95000 \times 100$$

$$\text{Loss \%} \Rightarrow 42750 / 95000 \times 100 = 45\% \text{ loss}$$

Sol-5

let S.P be Rs.100.

C.P will be Rs.60

Profit % = $40 / 60 \times 100 = 66.67\%$ approx.

Sol-6

Let C.P be Rs.100.

S.P will be Rs.65.

Loss% = $35 / 100 \times 100 = 35\%$ loss

Ans: 35% loss

Sol-7

S.P of an article = Rs 32,060

Loss = 30 %

70% of C.P = 32,060

C.P = $(32060 \times 100) / 70 = \text{Rs. } 45,800$

Now, Profit should be 10 %.

$\therefore \text{S.P} = 110\% \text{ of } 45,800 = \text{Rs. } 50,380$

Sol-8

Let CP of 1 article = Re.1

∴ CP of 35 article = Rs. 35

As per the question, CP of 35 article = SP of 28 article

∴ SP of 28 article = Rs. 35 & CP of 28 article = Rs. 28

∴ Profit % = $(35 - 28) / 28 \times 100 = 7 / 28 \times 100 = 25\%$

Shortcut Method:

Ask yourself whether profit or loss? If profit, profit % would be difference between number of articles divided by smaller number multiplied by 100 and if loss, loss % would be difference between number of articles divided by larger number multiplied by 100 will be your answer.

In the above question, there would be profit,

∴ Profit % = $(35 - 28) / 28 \times 100$

= $7 / 28 \times 100 = 25\%$

Sol-9

There would be profit,

∴ Profit % = $(42 - 35) / 35 \times 100$

= $7 / 35 \times 100 = 20\%$ profit

Sol-10

There would be loss,

∴ Profit % = $(40 - 32) / 40 \times 100$

= $8 / 40 \times 100 = 20\%$ loss

Sol-11

There would be loss,

$$\begin{aligned}\therefore \text{Loss \%} &= (25 - 15) / 25 \times 100 \\ &= 10/25 \times 100 = 40\% \text{ loss}\end{aligned}$$

Sol-12

Let the S.P be Rs. 100

$$C.P = 100/5 = \text{Rs.}20$$

$$\text{Profit \%} = 80/20 \times 100 = 400\%$$

Sol-13

$$S.P \text{ of the bike} = 120\% \text{ of } 72,500 = \text{Rs. } 87,000$$

$$S.P \text{ of the laptop} = 75\% \text{ of } 32,000 = \text{Rs. } 24,000$$

$$\text{Total S.P} = 87,000 + 24,000 = 1,11,000 \text{ \&}$$

$$\text{Total C.P} = 72,500 + 32,000 = 1,04,500$$

$$\text{Profit \%} = (111000 - 104500) / 104500 \times 100$$

$$\text{Profit \%} = (6500 / 104500) \times 100 = 6.22 \% \text{ profit approx.}$$

Sol-14

$$\text{Total investment} = \text{Rs.}20,00,000$$

$$\text{Total income} = \text{Rent} + S.P = (15,000 \times 5) + 23,00,000 = 75,000 + 23,00,000 = \text{Rs.}23,75,000$$

$$\text{Profit \%} = (2375000 - 2000000) / 2000000 \times 100$$

$$\text{Profit \%} = (375000 / 2000000) \times 100 = 18.75\%$$

Sol-15

C.P of 60 Kg apples = 4500 Rs.

S.P of 60 Kg apples = $80 \times 60 = \text{Rs. } 4800$

Profit % = $(300/4500) \times 100 = 6.67 \% \text{ approx.}$

Sol-16

Let S.P be $32x$

C.P be $28x$

Profit = $32x - 28x = 4x$

Profit: C.P = $4x : 28x = 1:7$

Sol-17

Cost price = $(5200 + 4500)/2 = 9700/2 = \text{Rs. } 4,850$

S.P = 115% of 4,850

S.P = $4850 \times 115 / 100 = \text{Rs. } 5,577.50$

Sol-18

To make the CP and SP of same number of articles, we will take LCM of 18 and 12 i.e. 36 Given,

CP of 18 glasses = Rs. 12

SP of 12 glasses = Rs. 18

CP of 36 article = Rs. 24

SP of 36 article = Rs. 54

Profit % = $(54 - 24) / 24 \times 100$

Profit % = $(30 \times 100) / 24 = 125\%$

Sol-19

To make the CP and SP of constant number of articles, we will take LCM of 30 and 25 i.e. 150

CP of 150 article = Rs. 125

SP of 150 article = Rs. 180

Profit of Rs 55 is earned by selling 150 boxes

$\therefore$ Profit of Re. 1 is earned by selling = $150 / 55$ boxes

$\therefore$ Profit of Rs. 770 is earned by selling = $(150 \times 770) / 55$

No of boxes = 2,100 boxes

Sol-20

Original profit = 30%

Let the original C.P = Rs. 100

S.P = Rs.130

, New C.P = Rs. 80

And New S.P = 120 % of 130 = Rs. 156

$\therefore$ New Profit % = $(76 / 80) \times 100 = 95\%$

% Change in profit = $(95-30)/30 \times 100$

$(65 / 30) \times 100 = 216.66\%$ approx.

Sol-21

To make the maximum possible profit, one has to purchase the toys at minimum rate and he must sell the same at maximum rate.

C.P. of 45 toys at minimum rate = $45 \times 175 = \text{Rs. } 7,875$ and

S.P. of 45 toys at maximum rate = $45 \times 300 = \text{Rs. } 13,500$

$\therefore$ Maximum possible profit = $13,500 - 7,875 = \text{Rs. } 5,625$

Sol-22

Let SP of 1 article = Re. 1

S.P of 20 article = Rs. 20 and

Profit = Rs. 8

C.P = Rs.12

Profit % = $(8 \times 100) / 12 = 66.66\%$ approx.

Sol-23

C.P of 1st item = $(2800 \times 100) / 140 = \text{Rs. } 2,000$

C.P of 2nd item = $(2800 \times 100) / 70 = \text{Rs. } 4,000$

Total C.P = $2000 + 4000 = \text{Rs. } 6,000$

Total S.P = Rs. 5,600

Loss = $6,000 - 5,600 = \text{Rs. } 400$

Sol-24

Sol: Let C.P of the article = Rs. x

Actual S.P = 95 % of x

assumed S.P = 120% of x

Since, difference between the two selling prices is Rs. 350

$$(120 x/100) - (95 x/100) = 350$$

$$(120 x - 95 x)/100 = 350$$

$$\Rightarrow 25 x = 35,000$$

$$x = 35000/25$$

$$x = 1,400$$

$$C.P = Rs. 1,400$$

$$Actual S.P. = 95\% \text{ of } 1,400 = Rs. 1,330$$

$$Actual S.P = Rs. 1,330$$

Sol-25

Let C.P of the article = Rs. x

Actual S.P = 80 % of x

assumed S.P = 140% of x

Since, difference between the two selling prices is Rs. 420

$$(140 x/100) - (80 x/100) = 420$$

$$(140 x - 80 x)/100 = 420$$

$$60 x = 42,000$$

$$x = 42000 / 60$$

$$x = 700$$

$$C.P = Rs. 700$$

$$S.P = 80 \% \text{ of } 700 = Rs. 560$$

Sol-26

$$S.P \text{ of 1 article} = 30/6 = \text{Rs. } 5$$

$$\text{Profit} = 25\%$$

$$\therefore C.P \text{ of 1 article} = 5 \times 100/125 = \text{Rs. } 4$$

$$\therefore C.P \text{ of 14 articles} = 4 \times 14 = \text{Rs. } 56$$

$$S.P \text{ of 14 articles} = \text{Rs. } 61.60$$

$$\therefore \text{Profit \%} = (61.60 - 56) / 56 \times 100 = 10\% \text{ profit}$$

Sol-27

$$C.P \text{ of 1st item} = (250 \times 100) / 110 = \text{Rs. } 227.27$$

$$C.P \text{ of 2nd item} = (250 \times 100) / 90 = \text{Rs. } 277.78$$

$$\text{Total C.P} = \text{Rs. } 505.05$$

$$\text{And Total S.P} = \text{Rs. } 500$$

$$\text{Loss} = 505.05 - 500$$

$$\text{Loss} = \text{Rs. } 5.05 \text{ approx.}$$

Sol-28

$$C.P \text{ of the printer} = \text{Rs. } x$$

$$C.P \text{ of the computer} = 160\% \text{ of } x$$

$$x + 1.6x = 31,200$$

$$2.6x = 31,200$$

$$x = 31200 / 2.6 = 12000$$

$$\text{Reqd. diff.} = 1.6x - x$$

$$.6x = .6 \times 12,000 = \text{Rs. } 7,200$$

Sol-29

Let C.P. of 1 gm = Rs.1

C.P of 850 gm = Rs. 850 and

S.P of 850 gm = Rs. 1,000 (since he is giving only 850 gm instead of 1000 gm)

Profit = Rs.150

% profit = $150/850 \times 100 = 17.65\%$ approx.

Sol-30

C.P of 900 gm = Rs 900

S.P of 900 gm = Rs 1300

$\therefore$ % profit = $(400/900) \times 100 = 44.44$ approx.

Sol-31

C.P of 900 gm = Rs 900

S.P of 900 gm = Rs 1100

% profit = $(200/900) \times 100 = 22.22\%$ approx.

Sol-32

C.P of 1200 gm = Rs 1000

CP of 900 gm = Rs 750

SP of 900 gm = Rs 1000

% profit = $250/750 \times 100 = 33.33\%$ approx.

Sol-33

$$\text{Rate of discount} = (22500 - 18000) / 22500 \times 100$$

$$\text{Rate of discount} = 4500/22500 \times 100 = 20\%$$

Sol-34

Let Marked price = Rs. 100

$$\therefore \text{Price after both the rebates} = (100 \times 70 \times 60) / (100 \times 100) = \text{Rs. } 42$$

$$\therefore \text{Single reduction} = 100 - 42 = 58\%$$

Sol-35

Let C.P = Rs. 100

Marked Price = Rs.120

$$\text{And S.P} = 120 \times 90/100 = \text{Rs. } 108$$

$$\text{Profit \%} = 8 \%$$

Sol-36

Let the original price = Rs. x

$$\therefore x \times 70\% \times 80\% = 56,000$$

$$\Rightarrow x = 56000 \times 100 \times 100 / 70 \times 80 = \text{Rs. } 1,00,000$$

Ans: Rs 1,00,000

Sol-37

Let the original price = Rs. x

$$x \times 86\% \times 92\% = 31,648$$

$$\Rightarrow x = (31648 \times 100 \times 100) / (86 \times 92) = \text{Rs. } 40,000$$

Sol-38

$$S.P = 36000 \times (85/100) \times (90/100) \times (95/100) = \text{Rs. } 26,163$$

Sol-39

Let the labeled price = Rs. x

$$x \times 85\% \times 75\% = 7,650$$

$$x = (7650 \times 100 \times 100) / (85 \times 75) = \text{Rs. } 12,000$$

Sol-40

$$\text{List Price} = 2250 \times (100/90) = \text{Rs. } 2,500$$

$$\therefore S.P = 2,500 \times (80/100) = \text{Rs. } 2,000$$

Ans: Rs 2,000

Sol-41

$$S.P = (3,000 \times 85\% \times 80\%)$$

$$S.P = 3,000 \times (85/100) \times (80/100) = \text{Rs. } 2,040$$

Sol-42

Let the first discount % = $(100 - x) \%$

$$2,0000 \times 80\% \times x\% = 12,000$$

$$x\% = (12000 \times 100 \times 100) / (80 \times 2000) = 75\%$$

$$1st\ discount = 100 - 75 = 25\%$$

Sol-43

$$S.P = 15,000 \times (90/100) \times (80/100) \times (75/100)$$

$$S.P = Rs. 8,100$$

Sol-44

Let the C.P = Rs.100

$$M.P = Rs. 160$$

$$S.P = (160 \times 85) / 100 = Rs.136$$

$$Profit \% = 36\%$$

Sol-45

$$S.P = 1,80,000 \times (97/100) + 1,70,000 \times (95/100)$$

$$S.P = 1,74,600 + 1,61,500 = Rs. 3,36,100$$

Sol-46

$$S.P \text{ of the Car} = 5,50,000 \times (80/100) \times (60/100) = \text{Rs. } 2,64,000$$

Ans: Rs 2,64,000

Sol-47

Let the marked price = Rs. x

Actual Cost Price = 85% of $x = 80x/100$

Assumed Cost Price = 80 % of $x = 80x/100$

$$(85x/100) - (80x/100) = 500$$

$$5x = 50,000$$

$$x = 10,000$$

Actual Cost Price = 85 % of 10,000 = Rs. 8,500

Sol-48

Let the marked price = Rs. 100

Selling Price = Rs. 80

Profit % = 25 %

$$C.P = (80 \times 100)/125 = \text{Rs. } 64$$

Without discount. S.P would be Rs. 100

$$\% \text{ profit} = (36/64) \times 100 = 56.25 \% \text{ approx.}$$

Sol-49

$$S.P = 85\% \text{ of } 30,000 = 25,500$$

$$\text{Profit} = 20\%$$

$$C.P = (25500 \times 100)/120 = 21,250$$

Since mixer grinder worth Rs. 1,500 is also given,

$$C.P \text{ of the refrigerator} = 21,250 - 1,500 = \text{Rs. } 19,750$$

Sol-50

$$S.P = 90\% \text{ of } 6,000 = \text{Rs. } 5,400$$

$$\text{Profit} = 25\%$$

$$C.P = (5400 \times 100)/125 = 4,320$$

Since pen drive worth Rs. 600 is also given,

$$C.P \text{ of the mobile phone} = 4,320 - 600 = \text{Rs. } 3,720.$$